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Q1.(Creating Dictionaries)

```
empty_dict1 = {}
```

```
empty_dict2 = dict()
```

```
print("Empty Dictionary (Method 1):", empty_dict1)
```

```
print("Type:", type(empty_dict1))
```

```
print("Empty Dictionary (Method 2):", empty_dict2)
```

```
print("Type:", type(empty_dict2))
```

```
student = {
```

```
    "name": "Vaishnavi",
```

```
    "city": "Latur",
```

```
    "course": "Polytechnic"
```

```
}
```

```
print("\nDictionary with String Keys:")
```

```
print(student)
```

```
print("Type:", type(student))
```

```
numbers = {
```

```
    1: "One",
```

```
    2: "Two",
```

```
    3: "Three"
```

```
}
```

```
print("\nDictionary with Integer Keys:")
```

```
print(numbers)
```

```
print("Type:", type(numbers))
```

```
mixed = {
```

```
    "name": "Vaishnavi", # String
```

```
"age": 18,      # Integer
"percentage": 89.5  # Float
}

print("\nMixed Data Type Dictionary:")

print(mixed)

print("Type:", type(mixed))
```

Q2.(Accessing and Modifying Elements)

```
student = {
    "name": "Vaishnavi",
    "age": 18,
    "city": "Latur",
    "marks": 85
}

print("Name:", student["name"])

student["marks"] = 92

student["grade"] = "A"

print("\nUpdated Dictionary:")

print(student)
```

Q3.(get(), keys(), values(), items())

```
person = {
    "name": "Priya",
```

```
"age": 21,  
    "profession": "Engineer"  
}  
print("Age:", person.get("age"))  
print("Salary:", person.get("salary", "Not Available"))  
print("\nKeys:")  
print(person.keys())  
print("\nValues:")  
print(person.values())  
print("\nKey-Value Pairs:")  
print(person.items())
```

Q4.(Nested Dictionary)

```
students = {  
    "s1": {  
        "name": "Vaishnavi",  
        "age": 20,  
        "marks": 88  
    },  
    "s2": {  
        "name": "Sneha",  
        "age": 21,  
        "marks": 95  
    }  
}
```

```
    }  
}  
  
print("First Student Details:")  
  
print(students["s1"])  
  
print("\nSecond Student Marks:")  
  
print(students["s2"]["marks"])  
  
students["s1"]["math"] = 90  
  
print("\nUpdated First Student Details:")  
  
print(students["s1"])
```

Q5.(update() and pop())

```
info = {  
    "brand": "Samsung",  
    "model": "A52",  
    "price": 25000  
}  
  
info.update({  
    "color": "Black",  
    "price": 24000})  
  
removed_value = info.pop("model")  
  
print("Removed Value:", removed_value)  
  
print("\nFinal Dictionary:")  
  
print(info)
```

Q6.(popitem() and clear())

```
student = {  
    "name": "vaishnavi",  
    "age": 20,  
    "city": "Pune",  
    "course": "Python",  
    "marks": 88  
}  
  
print("Original Dictionary:")  
print(student)  
  
removed1 = student.popitem()  
print("\nFirst Removed Item:", removed1)  
  
removed2 = student.popitem()  
print("Second Removed Item:", removed2)  
  
student.clear()  
  
print("\nFinal Dictionary after clear():")  
print(student)
```

Q7.(copy() and setdefault())

```
d = {  
    "a": 1,  
    "b": 2  
}
```

```
copy_dict = d.copy()
d.setdefault("c", 3)
d.setdefault("a", 100)
print("Original Dictionary:")
print(d)
print("\nCopied Dictionary:")
print(copy_dict)
```

Q8.(fromkeys() and Membership)

```
keys = ["name", "age", "city"]
person = dict.fromkeys(keys, None)
person["name"] = input("Enter your name: ")
person["age"] = int(input("Enter your age: "))
person["city"] = input("Enter your city: ")
print("\nDictionary:")
print(person)
search_key = input("\nEnter a key to check: ")
if search_key in person:
    print(search_key, "exists in the dictionary.")
else:
    print(search_key, "does not exist in the dictionary.")
```

Q9.(Practical Application)

```
contacts = {}

name = input("Enter Name: ")
phone = input("Enter Phone Number: ")
email = input("Enter Email: ")

contacts[name] = {
    "Phone": phone,
    "Email": email
}

search_name = input("\nEnter name to search: ")
contact = contacts.get(search_name)

if contact:
    print("\nContact Found:")
    print("Phone:", contact["Phone"])
    print("Email:", contact["Email"])
else:
    print("\nContact not found.")

print("\nAll Contacts:")

for name, details in contacts.items():
    print("Name:", name)
    print("Phone:", details["Phone"])
    print("Email:", details["Email"])
    print("-----")
```


Q10.(Mini Project - Combined Concepts)

```
students = {}

while True:

    print("\n===== Student Management System =====")

    print("1. Add New Student")

    print("2. Update Student Marks")

    print("3. Search Student")

    print("4. Display All Students")

    print("5. Remove Student")

    print("6. Exit")

    choice = input("Enter your choice (1-6): ")

    if choice == "1":

        roll = input("Enter Roll Number: ")

        name = input("Enter Name: ")

        age = int(input("Enter Age: "))

        marks = float(input("Enter Marks: "))

        students[roll] = {

            "Name": name,

            "Age": age,

            "Marks": marks

        }

        print("Student Added Successfully!")

    elif choice == "2":

        roll = input("Enter Roll Number: ")

        student = students.get(roll)
```

```
if student:

    new_marks = float(input("Enter New Marks: "))
student.update({"Marks": new_marks})

    print("Marks Updated Successfully!")

else:

    print("Student Not Found!")

elif choice == "3":

    roll = input("Enter Roll Number: ")
student = students.get(roll)

    if student:

        print("\nStudent Details")

        print("Name :", student["Name"])

        print("Age :", student["Age"])

        print("Marks:", student["Marks"])

    else:

        print("Student Not Found!")

elif choice == "4":

    if len(students) == 0:

        print("No Student Records Available.")

    else:

        print("\n----- Student Records -----")

        for roll, details in students.items():

            print("Roll No :", roll)

            print("Name  :", details["Name"])

            print("Age   :", details["Age"])
```

```
        print("Marks :", details["Marks"])

        print("-----")

    elif choice == "5":

        roll = input("Enter Roll Number: ")

        removed = students.pop(roll, None)

    if removed:

        print("Student Removed Successfully!")

    else:

        print("Student Not Found!")

    elif choice == "6":

        print("Thank You! Program Closed.")

        break

    else:

        print("Invalid Choice! Please Enter 1 to 6.")
```